Shobhit Dixit

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SUMMARY

AI/ML Engineer and Data Engineer with 4+ years of experience building and deploying production-grade machine learning systems. Expertise in NLP, Large Language Models, RAG architectures, computer vision, and end-to-end MLOps pipelines. Skilled in building end-to-end ML systems, from data ingestion and feature engineering through model training, evaluation, and cloud deployment

EXPERIENCE

ML Engineer (NLP Research) **Sep 2025 – Mar 2026**
AI/ML Co-op, Computational Social Science Lab, DU Philadelphia, PA

- Engineered scalable Python NLP pipelines using **BERT-based transformer models** for toxicity and sentiment classification on a 500K+ Reddit post dataset, achieving 89% F1-score and enabling quantitative community-health research
- Built end-to-end ML data pipelines with **Requests and BeautifulSoup** to automate web scraping, cleaning, and structuring of Reddit data, reducing manual data prep time by 70%
- Designed and extracted 20+ behavioral and temporal features (retention, engagement depth, contributor distribution) using **Pandas and NumPy**, improving downstream model predictive accuracy
- Validated model robustness via correlation analysis, VIF, and k-fold cross-validation, **reducing multicollinearity by 24%** and eliminating overfitting across 3 model architectures

Software Engineer (ML-Focused Platform Engineer) **Oct 2021 – Dec 2024**
Mobileum Bengaluru, India

- Designed and deployed Python/SQL **anomaly detection pipelines** using statistical thresholds over large-scale telecom system logs, **reducing network congestion by 22-24%** across 5+ EMEA enterprise accounts
- Automated log analysis and system monitoring workflows using **Python and Bash** scripting, enabling early anomaly detection and **cutting system downtime by 18–22%** through improved observability
- Integrated backend services (**Java/C++ APIs**) with real-time data workflows, **reducing detection latency by 16-20%** and enabling production-grade monitoring analytics for telecom clients
- Executed unit, system, and integration testing with comprehensive root-cause analysis across OS, network, and application layers, **reducing post-deployment incidents by 15-20%** and achieving 98% SLA resolution rate
- Collaborated cross-functionally with clients and engineering teams to translate complex network requirements into scalable technical designs, ensuring reliable delivery across global telecom accounts

PROJECTS

Enterprise AI Knowledge Assistant (LLM + RAG Platform) | Technologies: LangChain, OpenAI API, Pinecone, FastAPI, AWS [Link]

- Architected an enterprise RAG pipeline using **LangChain + OpenAI GPT** with **Pinecone vector search**, enabling semantic retrieval across 10,000+ enterprise documents with sub-second query latency
- Containerized and deployed inference services on **Docker + AWS EC2**, achieving scalable conversational AI workflows with 99.5% uptime

Credit Card Fraud Detection (ML + Cost Optimization) | Technologies: Python, Scikit-learn, SMOTE, Random Forest [Link]

- Boosted fraud recall from **67% to 87%** by engineering class-balancing pipelines using **SMOTE/ADASYN** on a 284K-transaction imbalanced dataset, achieving 95% precision and 0.86 F1-score
- Built a cost-benefit threshold optimization framework that **mitigated $10,000+ in estimated fraud losses** by tuning prediction cutoffs for real-world deployment

Intelligent Document Processing System | Technologies: PyTorch, Transformers, OCR, Python [Link]

- Developed a transformer-based NLP pipeline for financial document classification, achieving **91% accuracy** and reducing manual processing time by 60% through automated entity extraction and feature engineering

TECHNICAL SKILLS

ML/AI Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace Transformers, LangChain, OpenAI API
ML Domains: NLP, LLMs, RAG, Generative AI, Computer Vision, Feature Engineering, Model Evaluation, MLOps, ML Pipelines
Data & Analytics: Pandas, NumPy, EDA, Statistical Analysis, A/B Testing, Hypothesis Testing, ETL Pipelines, Web Scraping
Cloud & MLOps: AWS (EC2, S3, RDS), Azure, Docker, CI/CD, Git/GitHub, FastAPI
Programming: Python, SQL, C/C++, Bash, JavaScript/TypeScript
Databases: PostgreSQL, MS SQL Server, Oracle, BigQuery, Pinecone (Vector DB)
Tools: Linux/Unix, Windows, JIRA, Power BI, Jupyter, Azure DevOps

EDUCATION

Master of Science in Artificial Intelligence and Machine Learning (GPA: 3.93/4.0) | Drexel University, PA, USA **Mar 2026**
Bachelor of Technology in Information Technology | Krishna Institute of Engineering and Technology, Ghaziabad, India **Jul 2021**